MIL H-9884E
28 December 1984
SUPERSEDING
MIL-H-9884D
26 November 1973

MILITARY SPECIFICATION

HONEYCOMB MATERIAL, CUSHIONING, PAPER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This document covers a paper honeycomb structure which is used primarily as an impact energy dissipater for airdrop operations.

1.2 Classification. The paper honeycomb structure shall be one of the following classes and sizes, as specified (see 6.2).

Class 3 - Expanded and double faced honeycomb panel.
Size I - 36 inches by 48 inches
Size. II - 36 inches by 96 inches
Size III - 48 inches by 48 inches
Size IV - 48 inches by 96 inches

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1670

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SPECIFICATION

FEDERAL

MMM-A-180	Adhesive, Polyvinyl Acetate Emulsion
PPP-B-26	Bag, Plastic, (General Purpose)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-1163	Box, Corrugated Fiberboard, High Compression Strength, "Weather-Resistant, Wax Resin Impregnated
PPP-F-320	Fiberboard; Corrugated and Solid. Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-L-10547	Liners, Case, and Sheet, Overwrap: Water-Vaporproof or Waterproof, Flexible

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

Standard T 401 M	Fiber Analysis of Paper and Paperboard
Standard T 441, 05-63	Water Adsorptiveness of Non-bibulous Paper and Paperboard (Cobb Test)

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York, N.Y.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for esting and materials, 1916 Race Street, Philadelphia PA 19103.)

(Technical society and technical association documents are generally vailable for reference from libraries. They are also distributed among echnical groups and using Federal agencies

2.3 Order of precedence. In the event of a conflict between the text of his document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials (see 6.4).

3.2.1 Paper. The paper shall be at least 95 percent unbleached kraft fibers. If reclaimed fibers are used, the paper may contain traces of fibers other than unbleached kraft (not exceeding 5 percent). The maximum use of reclaimed fibers is encouraged, provided that the other requirements of this document are met. The maximum water absorptiveness (Cobb) of the paper shall be 0.01 gram per square centimeter for each side at an exposure time of not less than 5 minutes when tested as specified in 4.4.1.1.

3.2.2 Adhesive. The core and facing adhesive shall be a water resistant type, conforming to MMM-A-180 and to the water resistant requirement when tested as specified in 4.4.1.1.3.

3.3 Construction.

3.3.1 Honeycomb core. The honeycomb core shall be made of the paper specified in 3.2.1 which have been laminated by glue lines applied parallel to the machine direction of the paper. The nominal width of the glue lines shall be 3/16 inch. The center to center spacing of consecutive glue lines shall be 1 inch, nominal. Where increments are laminated together to form a honeycomb core, the offset in height between any two adjoining increments shall not exceed 1/16 inch.

3.3.2 Honeycomb panel. The honeycomb panel consists of honeycomb core which has been expanded and faced on each face with kraft facing paper specified in 3.2.1. The honeycomb core shall be made as specified in 3.3.1.

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The facing sheets shall be laminated to the expanded honeycomb core by glue and pressure and or heat applied to effect bonding. The honeycomb panel shall be furnished in the specified size with a tolerance of ± 1/2 inch (see 6.2). The thickness of the honeycomb panel shall be 3 ± 1/16 inches.

3.4 Average dynamic crushing stress.

3.4.1 Honeycomb panel. The average dynamic crushing stress of the honeycomb panel shall be 6300 ± 900 pounds per square foot to 70 percent strain when tested as specified in 4.4.3.3.

3.5 Percent of moisture.

3.5.1 Honeycomb panel. The percent of moisture of the honeycomb panel, not less than 24 hours after manufacture, shall not exceed 12 percent when tested as specified in 4.4.3.2.

3.6 Workmanship. The quality of the material shall conform to the quality established by this document.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractar is responsible for the performance of ail inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4,3 First article inspection. When a first article is required (see 6.2), it shall be made as specified below of samples applicable to the material being procured.

a. Class 3. One honeycomb panel (see 3.3.2), of the size specified in the contract or purchase order, for inspection as specified in 4.4.2.2, 4.4.2.3, 4.4.3.1, 4.4.3.2, and 4.4.3.3.

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4,4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Testing of paper.

4.4. 1.1.1 Lot and sample size. Unless otherwise authorized by the contracting activity (see 6.2), the lot size shall not exceed 100 tons {approximately 40 rolls). The sample size shall be five rolls. Specimens of the paper shall be selected at random from these rolls for the following tests:

4.4.1.1.2 Fiber identification and quantitative determination. Specimens shall be tested to determine that the paper meets the requirements specified in 3.2.1. Testing shall be accomplished in accordance with TAPPI Standard T 40l-M.

4.4.1.1.3 Water adsorptiveness. Specimens shall be tested to determine that the water absorptivenss for each side does not exceed the requirements specified in 3.2.1. Testing shall be accomplished in accordance with TAPPI Standard T 441, 0S. Results for each exposure shall be determined to the nearest 0.01 gram and the average of the 10 results calculated for each side.

4.4.2 End item inspection.

4.4,2,1 Inspection lot. The inspection lot shall consist of not more than 1000 honeycomb panels as applicable.

4.4.2.2 End item visual examination, class 3. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of panels. The sample unit shall be one panel. The inspection level shall be S–4 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Bonding of facing sheet	Any delamination of facing sheet from core greater than 2 inches in any direction.
Holes or crushed areas	Any hole or crushed area greater than 2 inches in any direction.
Core expansion	Uneven width of expanded core, resulting in facing sheet (paper) extending beyond core by more than 1/2 inch for a total distance of more than 6 inches.

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4.4.2.3 End item dimensional examination, class 3. The end item shall be examined for conformance to dimensions specified below. Any dimension not as specified shall be classified as a defect. The lot size shall be expressed in units of panels. The sample unit shall be one panel. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Length (specified ± 1/2	Oversize or undersize
Width (specified ± 1/2	Oversize or undersize
Thickness (3 ± 1/16)	Oversize or undersize

4.4.3 End item testing.

4.4.3.1 Water resistance, class 3. A specimen of honeycomb panel, 8 inches by 24 inches, shall be selected at random. Puncture the facing sheet, using a 3/8 dia tool, on both sides in not less than six places to permit water to enter the cell area when the speciment is immersed. Immerse the specimen in tap water at room temperature for not less than 60 minutes. The top of the specimen shall be not less than 1 inch below the surface of the water. After the 60 minute immersion period, the specimen shall be removed from the water and the excess water removed by shaking. Manually remove one facing sheet and examine for adhesive failure between the facing sheet and the honeycomb core. The adhesive used to laminate the facing sheet shall be unacceptable if adhesive failure occurs for a distance greater than 2 inches in any direction. Manually remove the other facing sheet and separate the paper plies across the width of the honeycomb core in three places. Examine the glue lines in the separations for adhesive failure. The adhesive used to laminate the plies shall be acceptable if more than one adhesive failure occurs in any one of the three separations.

4.4.3.2 Percent of moisture test, class 3. Two specimens measuring 16 inches by 18 inches, cut from diagonally opposite ends of each of five randomly chosen panels, in accordance with 4.4.3.3, shall be tested for the percent of moisture requirement as stated in 3.5.1 in accordance with 4.6.1.

4.4.3.3 Average dynamic crushing stress test, class 3. The sample unit shall consist of two specimens, each 16 inches by 18 inches, cut from each of five honeycomb panels selected at random from the lot. One specimen shall be cut from one end of the honeycomb panel; the second from the opposite end and side of the panel. The two specimens shall be bonded or taped to each other to form a 6-inch thick sample unit. Each sample unit shall be test for average dynamic crushing stress to 70 percent strain. The sample unit shall be impacted by a mass with a contact area of at least 16 inches by 18 inches at a velocity from 24 to 30 feet per second that will crush to at least 7

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percent of the sample thickness. The lot is acceptable if the crushing stress for each sample unit is 6300 ± 900 pounds per square foot to 70 percent strain. A retest for average dynamic crushing stress is permitted if four out of the five sample units tested conform to the requirement. The number of sample units for a retest shall be eight. The lot is acceptable if not less than seven out of the eight sample units tested conform to the 6300 ± 900 pounds per square foot average dynamic crushing stress requirement (see 6.6).

4.4.4 Packaging inspection. An examination shall be made to determine that the packing and marking complies with the section 5 requirements, Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 2.5 defects per 100 units.

Exam ine	Defect
Marking (exterior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as; incomplete closure of container flaps, loose strapping, improper taping or inadequate stapling. Bulged or distored container.

4.6 Methods of inspection.

4.6.1 Percent of moisture test. The percent of moisture may be obtained by subjecting the specimen to a temperature of approximately 360° Fahrenheit in a drying oven for a period of not less than 2 hours, except that when heat-sealable facing paper is used, the temperature shall be approximately 250° Fahrenheit for a period of not less than 4 hours. Each specimen shall be weighed to the nearest 1/4 ounce both prior to and after being dried. The percent of moisture shall be computed by dividing the loss in weight due to heating by the weight of the specimen befor heating, and multiplying by 100. The percent of moisture may also be obtained by a direct reading moisture meter equipped with a probe so that the percent of moisture is obtained from the expanded core. The length of the probe shall be not less than 12 inches in length. The probe shall be inserted in four widely dispersed points on the outer edges of the specimen. The maximum reading of the four results obtained shall be the value used to determine compliance with 3.5.1. The lot of expanded honeycomb panels shall be unacceptable if the percent of moisture in any specimen exceeds that specified in 3.5.1.

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5. PACKAGING

5.1 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.1.1 Level A packing.

5.1.1.1 Honeycomb panel (class 3). Eight honeycomb panels, of one size only, shall be packed in a snug-fitting shipping container conforming to overseas type, style A or I, grade A, type 2 load of PPP-B-601, or class 3, style optional, type 2 load of PPP-B-585. Each shipping container shall be waterproofed with a sealed case liner conforming to type I or II, grade C of MIL-L10547. Use of a sealed polyethylene plastic bag conforming to type I or II, grade A, style I, film thickness 0.006 inch of PPP-B–26 may be used in lieu of the MIL-L-10547 sealed case liner. Each container shall be closed and reinforced in accordance with the appendix of the applicable container specification. The weight and dimensional limitations of the container specification shall be waived. Skids are not required.

5.1.1.1. Individual packing. When specified (see 6.2), each class 3 honeycomb panel, prior to packing, shall be individually waterproofed in a clcse-fitting 0.006 inch thick polyethylene bag conforming to type I or II, grade A, style 1 of PPP-B-26. Prior to or during the bag closure operation, excess air within the bag shall be expelled. Shipping containers packed with polyethylene waterproofed honeycomb panels shall not require case liners.

5.1.2 Level B packing.

5.1.2.1 Honeycomb panel (class 3). Eight honeycomb panels of one size only, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be waterproofed with a sealed case liner conforming to type I or II, grade C of MIL-L-10547. Use of a sealed polyethylene plastiac bag conforming to type I or II, grade A, style I, film thickness 0.006 inch of O0O-B-26 may be used in lieu of MIL-L-10547 sealed case liner. Each container shall be closed and reinforced in accordance with the appendix of the applicable container specification. The weight and dimensional limitations of the container specification shall be waived. Skids are not required.

5.1.2.2 Weather-resistant fiberboard containers. When specified (see 6.2), the fiberboard shipping containers specified in 5.1.2.1 shall be grade V3c, V3s or V4s fiberboard boxes fabricated in accordance with PPP-B-636, or class I, type SWCFI, grade 275 of PPP-B-1163 and closed in accordance with method III as specified in the appendix of the container specification.

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5.1.3 Commercial packing. Honeycomb material packed as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.2 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The honeycomb panel is cut into sections and made into stacks which act as the energy dissipater for the landing shock of airdropped materiel.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. The class and size of honeycomb materiel required (see 1.2 and 3.3.2).
c. When a first article is required (see 3.1, 4.2, and 6.3).
d. selection of applicable level of packing (see 5.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.1.2.2).
f. When individual waterproofing of class 3 honeycomb panels is desired for level A shipments (see 5.1.1.1.1).
g. When the dynamic crushing stress test is not to be conducted at the U.S. Army Natick Research and Development Center (see 6.6).
h. When lot size is to exceed 100 tons (see 4.4.1.1.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample and consist of one unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.5 Moisture tester. The Hart Moisture Meter, West Islip, Long Island, New York, Type K-101 or Type K-103 with probe, is acceptable as a direct reading.

6.6 Crushing stress test. The average dynamic crushing stress test shall be accomplished at the U.S. Army Natick Research and Development Center, Airdrop Engineering Laboratory, Natick, MA 01760-5014, or at the Engineering Mechanics Research Laboratory, the University of Texas, Austin, TX. This testing may not be performed elsewhere unless engineering evaluation and prior approval of the testing facility is obtained from the contracting activity.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-9884E	Honeycomb Material, Cushioning, Paper

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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